SettleIQ

07_DATABASE_SCHEMA.md — Database Schema & Relationships

This document defines the minimum database structure required for the SettleIQ MVP. It supports financial file uploads, reconciliation runs, payment and settlement records, exceptions, AI-assisted investigation, human review, and audit logging.

1. Database Scope

The schema is intentionally MVP-focused. It should support the complete exception lifecycle without introducing unnecessary tables for features that are outside the approved scope.

Recommended database: PostgreSQL

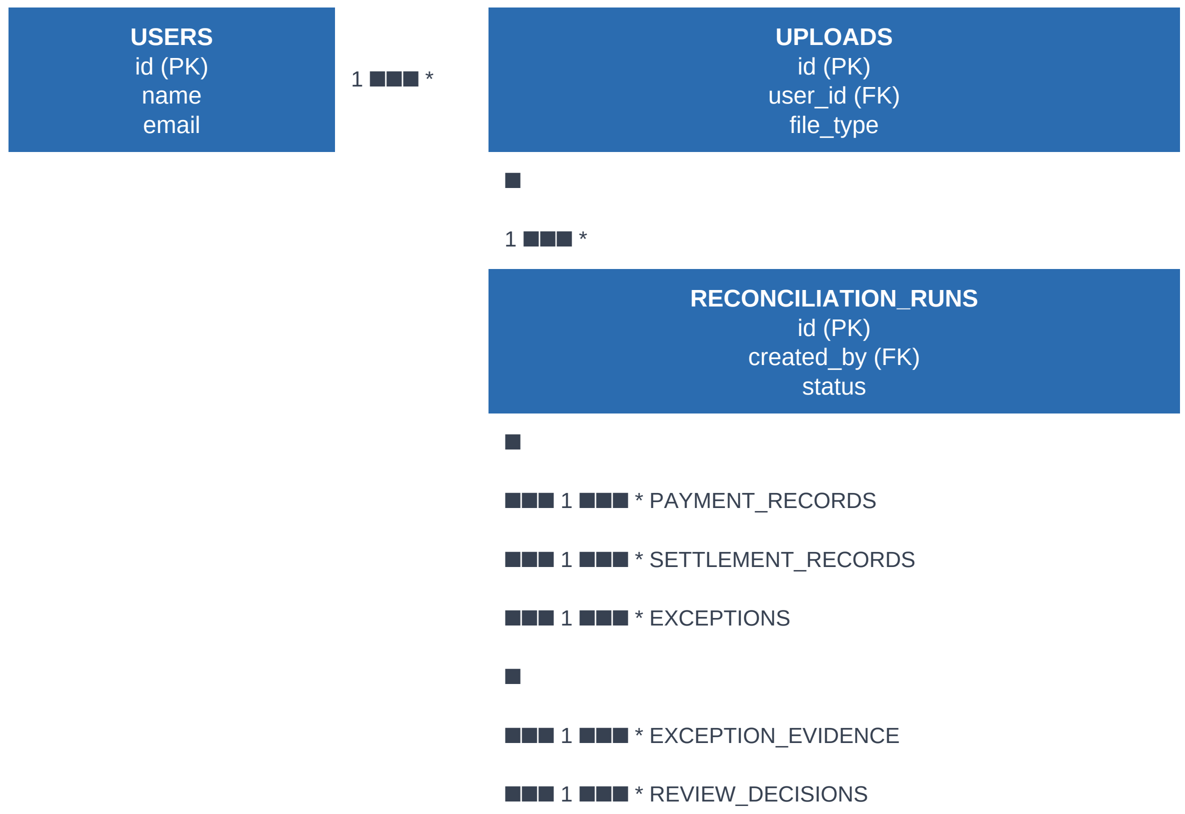
PostgreSQL is recommended because SettleIQ uses structured and related financial data and requires reliable relationships, queries, transactions, and auditability.

Core flow: User → Uploads → Reconciliation Run → Financial Records → Exceptions → Evidence / Decisions → Audit Log

2. Core Entities

Users — finance users interacting with the application.
Uploads — metadata for uploaded CSV files.
Reconciliation Runs — processing runs created from uploaded data.
Payment Records — normalized payment transactions.
Settlement Records — normalized settlement entries.
Exceptions — reconciliation issues identified by the system.
Exception Evidence — evidence gathered during investigation.
Review Decisions — guardrail or human-review outcomes.
Audit Logs — traceable record of important actions.

3. Entity Relationship Overview



4. Table Definitions

USERS

Stores the finance user responsible for uploads and actions.

Field	Purpose
id (PK)	Unique user identifier.
name	Display name.
email	Unique user email.
created_at	Account creation timestamp.

UPLOADS

Stores metadata about each uploaded financial file.

Field	Purpose
id (PK)	Unique upload identifier.
user_id (FK)	References the user who uploaded the file.
file_name	Original file name.
file_type	PAYMENTS or SETTLEMENTS.
storage_path	Location of the stored file.
status	UPLOADED, VALIDATED, or FAILED.
uploaded_at	Upload timestamp.

RECONCILIATION_RUNS

Represents one reconciliation execution.

Field	Purpose
id (PK)	Unique reconciliation run identifier.
created_by (FK)	References the user who started the run.
status	CREATED, PROCESSING, COMPLETED, or FAILED.
started_at	Processing start timestamp.
completed_at	Processing completion timestamp.

PAYMENT_RECORDS

Stores normalized payment records belonging to a reconciliation run.

Field	Purpose
id (PK)	Unique payment record identifier.
reconciliation_run_id (FK)	References the reconciliation run.
payment_reference	Source payment identifier.
amount	Payment amount.
currency	Currency code.
status	Source payment status.
transaction_date	Payment date/time.
raw_data	Original normalized source payload.

SETTLEMENT_RECORDS

Stores normalized settlement records belonging to a reconciliation run.

Field	Purpose
id (PK)	Unique settlement record identifier.
reconciliation_run_id (FK)	References the reconciliation run.
settlement_reference	Source settlement identifier.
amount	Settlement amount.
currency	Currency code.
settlement_date	Settlement date/time.
raw_data	Original normalized source payload.

EXCEPTIONS

Stores issues detected by the deterministic reconciliation engine.

Field	Purpose
id (PK)	Unique exception identifier.
reconciliation_run_id (FK)	References the run where the issue was found.
exception_type	Supported MVP exception category.
status	OPEN, INVESTIGATING, AUTO_RESOLVED, or HUMAN_REVIEW.
severity	Business priority.
source_reference	Relevant payment or settlement reference.
detected_at	Detection timestamp.

EXCEPTION_EVIDENCE

Stores evidence used to investigate and explain an exception.

Field	Purpose
id (PK)	Unique evidence identifier.
exception_id (FK)	References the related exception.
evidence_type	Type or source of evidence.
evidence_summary	Human-readable evidence summary.
confidence	Confidence value.
created_at	Evidence creation timestamp.

REVIEW_DECISIONS

Stores final guardrail or human-review outcomes.

Field	Purpose
id (PK)	Unique decision identifier.
exception_id (FK)	References the related exception.
recommended_action	Proposed resolution.
decision_outcome	AUTO_RESOLVE, HUMAN_REVIEW, APPROVED, or REJECTED.
confidence	Confidence associated with the recommendation.
decided_by	SYSTEM or USER reference.
reason	Explanation for the decision.
created_at	Decision timestamp.

5. Audit Logging

AUDIT_LOGS

Stores important user and system actions for traceability. Audit records should not be silently overwritten.

Field	Purpose
id (PK)	Unique audit event identifier.
user_id (FK, nullable)	User responsible for the action, when applicable.
action_type	Action such as FILE_UPLOADED or EXCEPTION_RESOLVED.
entity_type	Affected entity, such as UPLOAD or EXCEPTION.
entity_id	Identifier of the affected entity.
details	Relevant action details.
created_at	Event timestamp.

6. MVP Exception Types

Exception Type	Meaning
UNMATCHED_PAYMENT	A payment record cannot be matched to expected settlement information.
AMOUNT_MISMATCH	Related records exist but relevant amounts do not match.
DUPLICATE_RECORD	Potential duplicate payment or settlement record detected.
MISSING_SETTLEMENT	Expected settlement information is not present.
NEEDS_HUMAN_REVIEW	Evidence or confidence is insufficient for safe automatic resolution.

7. Key Database Rules

1. A reconciliation run must be traceable to its initiating user.
2. Every financial record belongs to one reconciliation run.
3. Every exception belongs to one reconciliation run.
4. AI-generated evidence must be stored separately from raw financial records.
5. AI must not directly modify financial records.
6. Resolution outcomes must be recorded through the decision workflow.
7. Important user and system actions must create an audit log entry.

8. Implementation Guidance

Build the schema incrementally. Start with Users, Uploads, Reconciliation Runs, Payment Records, Settlement Records, Exceptions, and Audit Logs. Add Exception Evidence and Review Decisions when the AI investigation and guardrail phases are implemented.

Scope rule: Do not introduce tables for Reports, Settings, standalone Q&A; / Ask AI, or other out-of-scope features unless explicitly requested.